

Common Ports and Protocols — Guided Notes ANSWER KEY

Unit 1.5.1 | CompTIA Network+ N10-009 Obj. 1.5 | N10-008 1.5

For instructor use / distribute after students complete the notes.

Question 1

Ports are _____ -bit numbers ranging from _____ to _____. The three ranges are:

Well-known: _____

Registered: _____

Dynamic/Ephemeral: _____

Answer: 16-bit; 0; 65535. Well-known: 0-1023; Registered: 1024-49151; Dynamic/Ephemeral: 49152-65535.

Question 2

Fill in the port numbers:

FTP control / data → _____ / _____

SSH → _____

Telnet → _____

SMTP → _____

DNS → _____

Answer: FTP: 21 (control) / 20 (data); SSH: 22; Telnet: 23; SMTP: 25; DNS: 53.

Question 3

Fill in the port numbers:

DHCP server / client → _____ / _____

HTTP → _____

HTTPS → _____

SMB → _____

RDP → _____

Answer: DHCP: 67 (server) / 68 (client); HTTP: 80; HTTPS: 443; SMB: 445; RDP: 3389.

Question 4

FTP transmits credentials in _____. Its secure replacements are _____ (runs over SSH, port 22) and _____ (FTP over TLS). Telnet is similarly insecure because _____.

Answer: Plaintext; SFTP; FTPS; all traffic including passwords travels unencrypted.

Question 5

DNS normally uses _____ (TCP/UDP) on port 53 for standard queries because _____. It switches to _____ for zone transfers and large responses because _____.

Answer: UDP; it's faster for small single request-response lookups; TCP; reliability is needed for larger transfers.

Question 6

When your browser connects to a web server on port 443, your OS assigns a temporary _____ port (range: _____) as the source port. This port exists only for the duration of that _____.

Answer: Ephemeral; 49152-65535; connection/session.

Question 7

SMB uses port _____. The _____ ransomware in 2017 exploited a vulnerability in SMB version _____, which is why modern best practice is to _____.

Answer: 445; WannaCry; SMBv1; disable SMBv1 entirely.

Question 8 — Real World Application

REAL WORLD: A user can browse the internet normally but cannot connect to the company's internal file server or map any network drives. You run netstat on the file server and see it listening on port 445. What is the most likely cause and how do you confirm it?

Answer: A firewall rule is blocking TCP port 445 between the user's subnet and the file server. Confirm by checking the firewall policy for rules affecting SMB/port 445. Test by attempting a connection with telnet or a port scanner (nmap) to the server on port 445 from the affected workstation — a timeout or connection refused confirms the block. Also check Windows Firewall on the server itself.